

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 311 — Past Questions & Materials
by Renance

Course: MTH 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 300

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Paper 1 of 3 — MTH 311 Abstract Algebra 11 2020 2021 First Semester

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION 2020 /2021

Course Code/ Title: MTH 311-Abstract Algebra II |

Instructions: Attempt any 4 questions. -Time Allowed: 25 Hours

Question 1: :

(a) What is a normal subgroup?

Let $\{(1), (12)\}$ be a subgroup of group $G = S_3$.

Show that K is a normal

subgroup of G . (6 marks).

(b) Let G be a group and N a

subgroup of G . 'Prove that N is normal in G if

and only if $g^{-1}ng \in N$ for all

$g \in G$ and $n \in N$. (6 marks). _

(c) Hence show that the alternating

group of degree n (A_n) is a normal subgroup

of the symmetric group of degree n (S_n).

(55 marks).

Question 2:

(ii) conjugate elements. Show that the set

of all conjugate elements of g is normal in G . (6 marks).

(a) Define: (i) conjugate subgroups

_ Of all conjugate elements of g

(c) Show that the center of the

Symmetric group S_n is trivial for $n = 3$. (45

marks),

Question 3:

(a) Define a quotient group. Prove that the set of all right cosets (left cosets)

of a normal subgroup K of a group G , defined in G/K as $gK * mK = (gm)K$

is a quotient group. (65 marks).

(b) What is an index of a subgroup H in a group G ? Let $G = S_3$ and

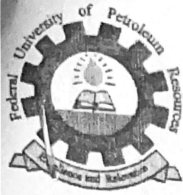
ii) the

$(1), (128), (132)\}$ a subgroup of G . Then find: (i) the index of H in G (ii) quotient group G/H . (6 marks)

Paper 2 of 3 — MTH 311 and 322 Abstract Algebra 11 2018 2019 First Semester

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
SECOND/FIRST SEMESTER EXAMINATION 2017/2018 2018/2019 SESSION
COURSE TITLE & CODE: ABSTRACT ALGEBRA II MTH 322/MTH 311

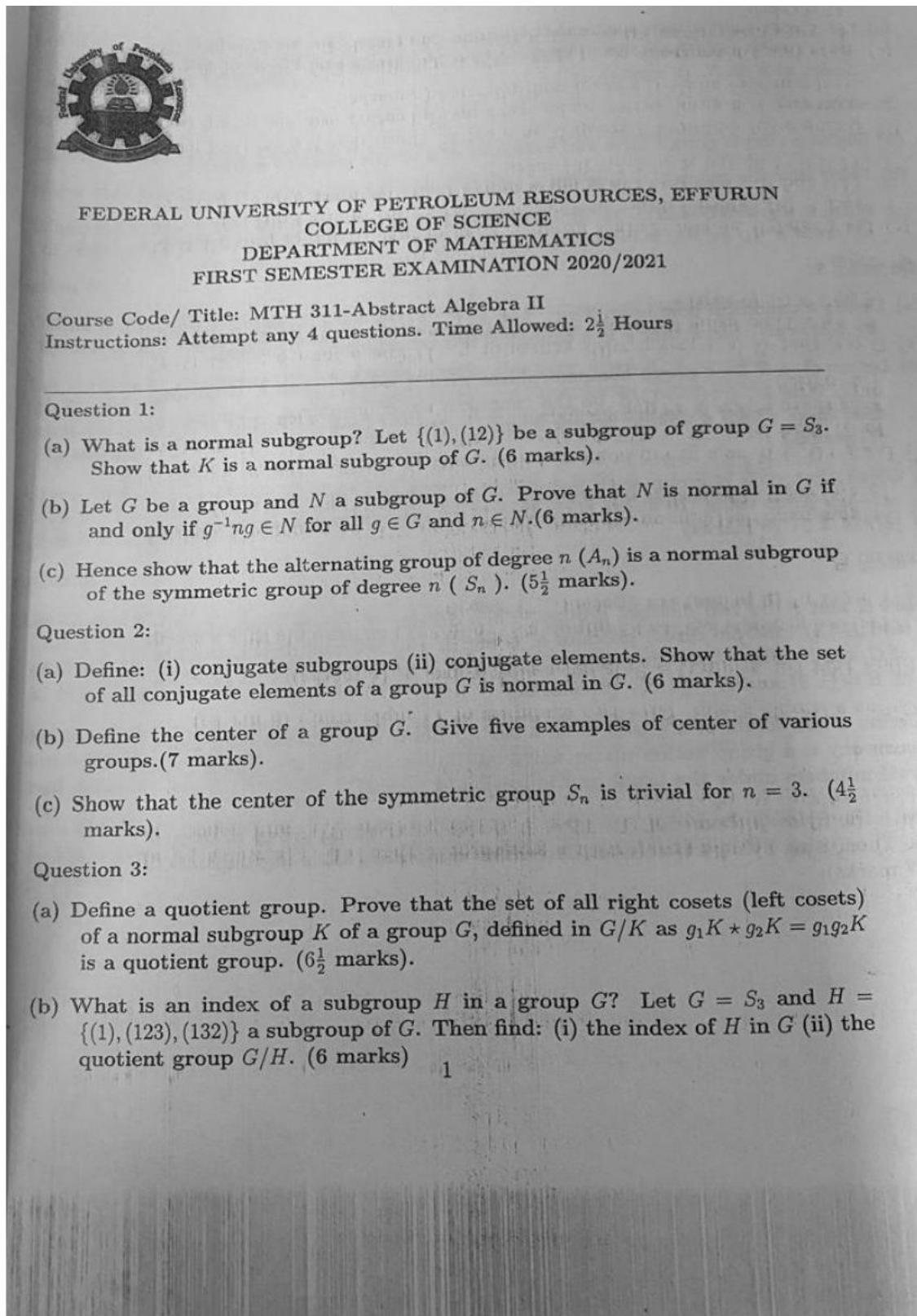
Instruction: Attempt any four questions Time: 2 hours

1. a. Define the following (i) Contents of a polynomial over $\mathbb{Z}$ (ii) primitive content (iii) Subfield and Field Extension
 b. (i) Let $f(x)$ and $g(x)$ be primitive polynomials. Prove that $f(x)g(x)$ is also a primitive polynomial (ii) What is the content of the following polynomials $3x^2 + 6x + 12$ and $x^5 + 3x^2 + 4x - 5$.
 c. Let p be a prime number. Show that $f(x) = x^{p-1} + x^{p-2} + \dots + x + 1$ is irreducible in $\mathbb{Z}[x]$
2. a. Define the following (i) Group homomorphism (ii) Group Isomorphism (iii) Cyclic subgroup
 b. In a group homomorphism, prove the following (i) $f(e_1) = e_2$, where e_1 is the identity of G_1 and e_2 the identity of G_2 . (ii) $f(x^{-1}) = [f(x)]^{-1}$ for all x in G_1 .
 c. Consider the groups $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ and define $f: (\mathbb{C}, +) \rightarrow (\mathbb{R}, +)$ by $f(x + iy) = x$, the real part of $x + iy$. Show that f is a homomorphism. What are Image of f and Kernel of f ?
3. a. Define the following (i) Even and Odd permutation (ii) Disjoint cycles (iii) Permutation of a symmetric group
 b. Given the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 4 & 7 & 2 & 1 & 3 & 6 & 5 \end{pmatrix}$ (i) Express it as a product of disjoint cycles (ii) Find its inverse (iii) Express it as a product of transpositions (iv) Verify if it commutes with the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 7 & 2 & 3 & 1 & 5 & 6 & 8 \end{pmatrix}$
 c. List out the even permutations of the symmetric group S_4
4. a. Define the following (i) External Direct product of groups (ii) Internal Direct product of groups (iii) State any of the sylow's theorem without proof
 b. (i) Show that a group G of order 30 either has a normal subgroup of order 5 or a normal subgroup of order 3 (ii) Show that $\mathbb{Z}_{10}$ is the Internal direct product of its subgroups $H = \{0, 5\}$ and $K = \{0, 2, 4, 6, 8\}$
 c. Let a group G be the internal direct product of its subgroups H and K then prove that each $x \in G$ can be uniquely expressed as $x = hk$ where $h \in H, k \in K$ and that $hk = kh \forall h \in H, k \in K$.
5. a. Define the following (i) Euclidean domain (ii) Principal Ideal domain (iii) Unique factorization domain
 b. (i) Show that $\mathbb{Z}$ (the set of all integers) is a Euclidean domain. (ii) Find all the units in $R = \{a + b\sqrt{-5}; a, b \in \mathbb{Z}\}$.
 c. What is the greatest common divisor of two integers (g.c.d.) and find the g.c.d. of the following (i) $p(x) = x^2 + 3x - 10$ and $q(x) = 6x^2 - 10x - 4$ (ii) 614 and 513
6. a. Define the following (i) Group (ii) Normal Subgroup (iii) Quotient group
 b. (i) Prove that every subgroup of a commutative group is normal
 (ii) Show that the subgroup $\langle (2 \ 3) \rangle$ of the Symmetric group S_3 is not normal
 c. Given the Quaternion group $Q_8 = \{\pm I, \pm A, \pm B, \pm C\}$ where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, C = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$ and $i = \sqrt{-1}$. Show that the cyclic subgroup $\langle A \rangle$ is normal in Q_8

Paper 3 of 3 — MTH 311, 2012

Level 300 • First Semester • PastQuestion

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- (c) Let $G = C_4$ be a cyclic group with a generator a , that is $G = \{e, a, a^2, a^3\}$ and $K = \{e, a^2\}$ a subgroup of G . Then find the quotient G/K and hence the index of K in G . (5 marks)

Question 4:

- (a) Define a simple group. Give two examples of a simple group (6 marks).
 (b) Show that the group $G = S_3$ is not a simple group. (4 marks)
 (c) When is a group G said to be nilpotent? Show with an example that a group of order p^n (p prime) is nilpotent. ($7\frac{1}{2}$ marks)

Question 5:

- (a) Define a canonical homomorphism. Let $\phi : G \rightarrow G/H$ be a mapping defined by $\phi(a) = gH$. Show that ϕ is a canonical homomorphism. (5 marks).
 (b) Let $\psi : G \rightarrow H$ be a group homomorphism with $K = \ker \psi$, then K is normal in G . And let $\phi : G \rightarrow G/K$ be the canonical homomorphism. Then prove that there exists a unique isomorphism $\eta : G/K \rightarrow \psi(G)$ such that $\psi = \eta\phi$. ($7\frac{1}{2}$ marks)
 (c) Given that G is a group with generator g . Define a map $\phi : \mathbb{Z} \rightarrow G$ by $n \mapsto g^n$. Then using (b) above show that there exists a unique isomorphism $\mathbb{Z}/m\mathbb{Z} \cong G$. (5 marks)

Question 6:

- (a) Let G and H be two distinct groups. Then define: (i) direct product of G and H (ii) external direct product of G and H (iii) internal direct product of G and H . Show that the group of units $U(8)$ is the internal direct product of $H = \{1, 3\}$ and $K = \{1, 5\}$. ($6\frac{1}{2}$ marks).
 (b) Define a left action of a group G on a set X . Show that the set $\mathbb{N}$ of natural numbers is a group action under natural multiplication over the group G of real numbers under the usual multiplication. (4 marks)
 (c) State the Sylow's theorems. Let $G = A_5$ be the alternating group of degree 5. Then using Sylow's theorems, determine completely the subgroups of G . (7 marks).